

Lecture 7 – Properties of the MLE

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In Lecture 6, we constructed estimators using the method of moments and maximum likelihood. We now finish our discussion of maximum likelihood and turn to the second question from Lecture 1: *how do we determine whether an estimator is good?*

7.1 Maximum likelihood for functions of a parameter

Suppose the quantity we want to estimate is $\eta = g(\theta)$. For example, in a Bernoulli model we might want to estimate p^2 or $p(1 - p)$ instead of p . To define an MLE for η , we distinguish two cases.

One-to-one transformations

If g is one-to-one on Θ , each $\eta \in g(\Theta)$ corresponds to exactly one parameter value $\theta = g^{-1}(\eta)$. Thus η gives a new label for the same distribution.

Definition 1 (MLE of a one-to-one function of the parameter). The likelihood for $\eta = g(\theta)$ is

$$L_{\underline{x}}^g(\eta) = L_{\underline{x}}(g^{-1}(\eta)), \quad \eta \in g(\Theta).$$

An **MLE of η** is any value $\hat{\eta} = \hat{\eta}(\underline{x})$ satisfying

$$L_{\underline{x}}^g(\hat{\eta}) = \max_{\eta \in g(\Theta)} L_{\underline{x}}^g(\eta),$$

provided the maximum is attained.

Transformations that are not one-to-one

If g is not one-to-one, several values of θ can give the same η . We assign η the largest likelihood available among parameter values compatible with it.

Definition 2 (MLE of a function that is not one-to-one). The **profile likelihood** for $\eta = g(\theta)$ is

$$L_{\underline{x}}^g(\eta) = \max_{\theta \in \Theta: g(\theta) = \eta} L_{\underline{x}}(\theta), \quad \eta \in g(\Theta).$$

An **MLE of η** is

$$\hat{\eta} = \arg \max_{\eta \in g(\Theta)} L_{\underline{x}}^g(\eta),$$

provided it exists and is unique.

Invariance of the MLE

With these definitions in place, there is a simple way to find an MLE of $g(\theta)$ from an MLE of θ .

Theorem 1 (Invariance of the MLE). If $\hat{\theta}$ maximizes the likelihood for θ , then $g(\hat{\theta})$ is an MLE for $g(\theta)$ under the definitions above.

Proof. Fix the observed sample $\underline{x}$. Since $\hat{\theta}$ is a global maximizer,

$$L_{\underline{x}}^g(\eta) \leq L_{\underline{x}}(\hat{\theta}) \quad \text{for every } \eta \in g(\Theta).$$

At $\eta = g(\hat{\theta})$, the supremum includes $\theta = \hat{\theta}$, so

$$L_{\underline{x}}^g(g(\hat{\theta})) = L_{\underline{x}}(\hat{\theta}).$$

Thus $g(\hat{\theta})$ maximizes $L_{\underline{x}}^g$. □

Example 1 (Estimating a squared probability). For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(p)$, $p \in [0, 1]$, Lecture 6 gave $\hat{p} = \bar{X}_n$. The function $g(p) = p^2$ is one-to-one on $[0, 1]$, so the MLE of $\eta = p^2$ is

$$\hat{\eta} = (\hat{p})^2 = (\bar{X}_n)^2.$$

Example 2 (Estimating a Bernoulli variance). For $X_i \stackrel{\text{iid}}{\sim} \text{Ber}(p)$, suppose the target is $\eta = p(1 - p)$, the variance of the Bernoulli. This function is not one-to-one: p and $1 - p$ give the same variance. Nevertheless, invariance gives

$$\hat{\eta} = \bar{X}_n(1 - \bar{X}_n).$$

It is unnecessary to solve separately for the two values of p compatible with each proposed variance.

7.2 MLEs and sufficient statistics

In Lectures 3–4, we showed that a sufficient statistic retains all the information about θ . Maximum likelihood gives one concrete consequence of this statement.

Theorem 2 (MLEs depend on sufficient statistics). Suppose T is sufficient for θ and there is a unique MLE $\hat{\theta}$. Then $\hat{\theta}$ depends only on $T(\underline{x})$.

Proof. By the Neyman–Fisher factorization theorem,

$$L_{\underline{x}}(\theta) = q(T(\underline{x}), \theta)h(\underline{x}).$$

But $h(\underline{x}) \geq 0$ and does not depend on θ , so

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L_{\underline{x}}(\theta) = \arg \max_{\theta \in \Theta} q(T(\underline{x}), \theta).$$

The right side depends on $\underline{x}$ only through $T(\underline{x})$. Hence $\hat{\theta}$ is a function of $T(\underline{x})$. □

Connection with exponential families and moment matching

For the one-parameter exponential family from Lecture 5,

$$f_{\theta}(x) = g(x) \exp\{b(\theta)R(x) - c(\theta)\},$$

the log likelihood for an independent sample is

$$\ell_{\underline{x}}(\theta) = \sum_{i=1}^n \log g(x_i) + b(\theta) \sum_{i=1}^n R(x_i) - nc(\theta).$$

The only part of the data needed for maximization is the sufficient statistic $\sum_i R(X_i)$. An interior MLE, when differentiability holds, satisfies

$$0 = b'(\hat{\theta}) \sum_{i=1}^n R(x_i) - nc'(\hat{\theta}) \implies \frac{1}{n} \sum_{i=1}^n R(x_i) = \frac{c'(\hat{\theta})}{b'(\hat{\theta})}.$$

Under the conditions for the moment identity in Lecture 5, the right side is $\mathbb{E}_{\hat{\theta}}[R(X)]$. Thus the likelihood equation matches the empirical mean of the natural statistic to its theoretical mean. Boundary values and whether a solution is a global maximum must still be checked.